

TERM 3 -
P.5 LESSON NOTES.

APPROVED: a) H.O.D
b) D/Academic

WK 1 and 2

Completion of work carried forward from 2nd term

WK. 3 LESSON 1

Time can be written either in the 12 hour clock system or the 24 hour clock system.

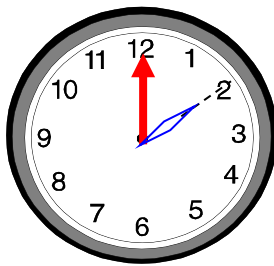
THE 12 HOUR CLOCK SYSTEM

In this system, we use a.m and p.m to tell the time. We use a.m to tell the time after midnight and in the morning, then p.m. in the afternoon and evening.

Telling time using a.m. and p.m.

Example.

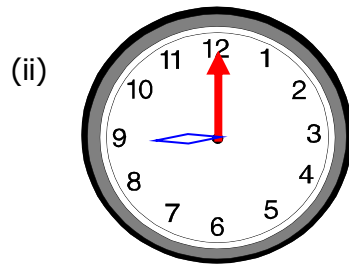
Write the time shown on the clock face in words and in figures.



After midnight.

It is 2 o'clock in the morning or 2:00 a.m.

NB. A NEW DAY BEGINS AT MIDNIGHT



What is the time?

The time is 9:00 p.m or 9 o'clock at night.

Activity

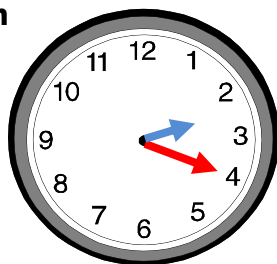
Mk. Book 5 pg 251

Macmillan Primary maths pg. 165 – 166

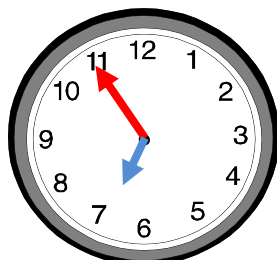
Write these items in words.

- a) 8:10am – Ten minutes past 8 in the morning.
- b) 6:00 am
- c) 3:52 pm
- d) 9:35 pm

e) Afternoon



f) Morning



WK. 3 LESSON 2

THE 24 HOUR CLOCK SYSTEM

In this system of telling time, we don't separate the time into am and pm. All the 24 hours of the day are written continuously using four digits ending with the word hours.

12 hour clock	24 hour clock
12:00 midnight	2400hrs
1:00 a.m	01 00 hrs
2:00 a.m	01 00 hrs
3:00 a.m	01 00 hrs
4:00 a.m	01 00 hrs
5:00 a.m	01 00 hrs
6:00 a.m	01 00 hrs
7:00 a.m	01 00 hrs
8:00 a.m	08 00hrs
9:00 a.m	09 00hrs
10:00 a.m	10 00hrs
11:00 a.m	11 00hrs
12:00 noon	12 00hrs

1:00 p.m	13 00 hrs
2:00 p.m	14 00 hrs
3:00 p.m	15 00 hrs
4:00 p.m	16 00 hrs
5:00 p.m	17 00 hrs
6:00 p.m	18 00 hrs
7:00 p.m	19 00 hrs
8:00 p.m	20 00 hrs
9:00 p.m	21 00 hrs
10:00 p.m	22 00 hrs
11:00 p.m	23 00 hrs
12:00	24 00 hr

**When the 12 hr clock time is am the hours in the 24 clock system do not change. Eg.
[1:00am = 0100hours] and [10:00am = 1000hours]**

Expressing 12 hour clock system to 24 hour time.

(a) Change 9:00am to 24 hour clock.

**am time does not change we only add 0000 hours

$$\begin{array}{r} 9:00 \\ + \\ 0000 \\ \hline 0900 \text{ hours} \end{array}$$

(b) Change 2:30am to 24 hours clock system

**

$$\begin{array}{r} 2:30 \\ + \\ 0000\text{hrs} \\ \hline 0230\text{hrs} \end{array}$$

(c) Change 7:00pm to 24 hour clock on pm time, we add 12hrs apart from 12:00pm which remains as 1200hrs.

**7:00pm

$$\begin{array}{r} + \\ 12: 00\text{hrs} \\ \hline 19 00\text{hrs} \end{array}$$

ACTIVITY

Express the following in 24 hours clock system

- (a) 3:00am
- (b) 9:00am
- (c) 2:00pm
- (d) 4:00pm
- (e) 12:00Midnight

CHANGING 24 CLOCK SYSTEM TO 12 HOUR CLOCK.

We subtract 12hours from any time above 1200hrs to change to 12 hour clock. When 12hrs are subtracted the time is then written as pm.

Example

(a) Change 1500hrs to 12 clock time.

$$\begin{array}{r} 1500\text{hrs} \\ -1200\text{hrs} \\ \hline 3:00\text{pm} \end{array}$$

(b) Write 2310 hours to a 12hrs clock system.

$$\begin{array}{r} 2310 \\ -1200 \\ \hline 11:10\text{pm} \end{array}$$

(c) Write 0200hrs to a 12hrs clock system.

$$\begin{array}{r} 0200 \\ - 0000 \\ \hline 2:00 \text{ am} \end{array}$$

N.B For time below 12 hours, we subtract 0000hrs to change to 12hrs clock system, this time then becomes am.

ACTIVITY

Change the following to 12hr clock system.

- (a) 1300hrs
- (b) 1205hrs
- (c) 0300hrs
- (d) Our lessons start at 0830hrs everyday. Every that time on a 12 hour clock.

WK 3 LESSON 3

DURATION OF TIME

Duration means amount of time taken between different intervals.

Finding Duration

1. Luyiga started walking from her house at 7:15am and reached the town at 9:15am.

How long did it take her?

Ending time = 9:15am

Starting time = 7:15am
2:00am

It took her 2 hours.

2. A forty minute lesson started at 3:30pm, at what time did it end.

$$\begin{array}{r} 3:30\text{pm} \quad 30 + 40 = 70 \\ 40 \quad 70/60 = 1 \text{ rem } 10 \\ \hline 4 \text{ } 10\text{pm} \end{array}$$

3. Nawumi went to the well to fetch waste. She left home at 3:35pm. She returned home half an hour later. At what time did she return home?

3:35

30

Reference: MK Book 5 pg 252, Macmillan PR MATHS BK 5 pg. 166

WK 3 : LESSON 4 & 5

INTERPRETING TIME TABLES

When interpreting time tables, it is important to identify the arrival time and departure time at a given place.

The table below shows a distance – time table for a bus travelling from Masindi to kitgum.

TOWN	DISTANCE FROM MASINDI	ARRIVAL	DEPARTURE
MASINDI			9:00am
KIGUMBA	39KM	9:40am	10:00am
KAMUDINI	115KM	11:10am	11:25am
LIRA	191KM	12:30pm	1:00pm
KITGUM	125KM	3:15pm	4:55pm

- (a) At what time did the bus arrive at Kamudini
- (b) What was the departure time of the bus from Lira?
- (c) What is the distance between Kigumba and Lira?
- (d) How long does the bus take to travel from Masindi to Kitgum?

Macmillan Primary 5 PG 168 – 169

WK . 3 LESSON 6

RELATION OF SPEED, DISTANCE AND TIME.

A. FINDING DISTANCE

1. Find the distance covered in 2 hours at a speed of 90km per hour.

NB. Distance means amount of length covered.

It is measured in kilometers (KM)

$$\begin{aligned}\text{Distance} &= \text{Speed} \times \text{Time} = S \times T \\ &= 90\text{km/hr} \times 2 \text{ hrs} \\ &= \underline{\underline{180\text{km}}}\end{aligned}$$

2. A lorry is travelling at 50km/hr. how far can it go in 3 hours?

$$\begin{aligned}D &= S \times T \\ &= 50\text{km/hr} \times 3 \text{ hrs} \\ &= \underline{\underline{50\text{km}}}\end{aligned}$$

WK. 4 LESSON 1 & 2

B. FINDING TIME

To find time we divide distance by speed.

- (i) Calculate the time taken by a car travelling at 60Kph to cover a distance of 480Km.

$$\text{Time} = \text{Distance} / \text{Speed}$$

$$= \frac{480\text{km}}{60\text{km/hr}}$$

$$= \underline{\underline{8 \text{ hours}}}$$

- (ii) A taxi travelled at a speed of 70km every hour to cover a distance of 245km. How long did it take?

$$T = \frac{D}{S}$$

$$= \frac{245\text{km}}{70\text{km/hr}}$$

$$= \frac{49}{14}$$

$$= \frac{7}{2}$$

$$= 3\frac{1}{2} \text{ hrs}$$

C. FINDING SPEED

Speed means the amount of distance covered per given time. To find speed. We divide the distance by the time taken.

$$\text{Therefore, Speed} = \frac{\text{Distance}}{\text{Time}}$$

$$S = \frac{D}{T}$$

The units for speed are Km per Hour (Km/hr)

Question

1. A bus covers a distance of 256Km in 4 hours. What speed was the bus travelling at?

$$S = \frac{D}{T} = \frac{256\text{km}}{4\text{hr}} = 64\text{Km/hr}$$

Ref: Macmillan Book 5 pg. 167 – 168, Mk Books.

WK. 4 LESSON 5 & 6

MONEY

CURRENCY, CONVERSION AND EXCHANGE RATES.

- **CURRENCY:** Is the name given to the type of money used in a given country.
- **EXCHANGE RATE :** This is the price for buying another country's currency. The Foreign Exchange (Forex Bureau) and the central Bank are responsible for buying and selling of foreign currency.
- People who travel to other countries need to exchange their local currency for currencies used in the countries they go.
- If you have Uganda currency, the forex bureau sells to your foreign currency.
- If you have foreign currency, the forex bureau buys it from you and gives you Uganda currency.

THE TABLE BELOW SHOWS THE EXCHANGE RATES OF DIFFERENT CURRENCIES IN UGANDA FOR A CERTAIN PERIOD AT CRANE FOREX BUREAU.

CURRENCY	BUYING	SELLING
1 pound sterling (£)	Ug. Sh. 2,500	Ug. Sh. 2,550
1 Us dollar (US \$)	Ug. Sh. 1,700	Ug. Sh. 1,720
1 Kenya shillings (Ksh)	Ug. Sh. 19	Ug. Sh. 20
1 Rwanda franc (R.F)	Ug. Sh. 1.9	Ug. Sh. 2.2
1 Euro (Eur)	Ug. Sh. 1,520	Ug. Sh. 1,560
1 Tanzania shillings (Tz-sh)	Ug. Sh. 1.6	Ug. Sh. 2

Using exchange rates to convert currencies.

Example:

A tourist arrived in Uganda with £7, 650. He had to convert it to Uganda shillings. The exchange rate in forex bureau is £1= Ug. Shs. 2,500.

The tourist has pounds so the bureau will buy from him.

£1 costs 2,500

Therefore

£7,650 will cost { sh. 2500 x 7650)

He gets shs. Shs. 19, 125,000

Example 2

Tamu is to travel to Germany with Euros equivalent to Ug. Shs. 12,480,000. Find the amount in euros Tamu will travel with. The exchange rate is 1 Euro = Ug. Shs 1560

$$\begin{aligned}\text{Therefore; Ug.shs. } 12,480,000 &= \frac{12,480,000}{1,560} \\ &= \underline{\underline{8,000 \text{ Euros}}}\end{aligned}$$

Ref: MK BK.6 Pg. 217 – 224

WK. 5 LESSON 1 & 2

TEMPERATURE

Temperature is the degree of hotness or coldness. We use thermometers to measure temperatures. Temperatures is measured in degrees.

Either Celsius (°C) or Fahrenheit.

Difference in temperature.

1. The temperature of water before heating was 30°C, after heating the temperature rose to 60°C, what was the rise in temperature?

$$\begin{aligned}\text{Rise in Temp} &= 60^{\circ}\text{C} - 30^{\circ}\text{C} \\ &= \underline{\underline{30^{\circ}\text{C}}}\end{aligned}$$

2. The temperature of food at the time of serving was 95°C after leaving it on the plate for 10 minutes the temperature was 48°. What was the fall in temperature?

$$\begin{aligned}\text{Fall in temp} &= \text{Temp after serving} - \text{Temp after 10 minutes.} \\ &= 95^{\circ} - 48^{\circ} \\ &= \underline{\underline{47^{\circ}\text{C}}}\end{aligned}$$

3. The temperature of a bottle of soda from a fridge was -4°C . After 15 mins it became 10°C what was the rise in temperature?

$$\text{Rise in temp} = 10 - (-4) = 10 + 4 = 14^{\circ}\text{C as the rise in Temp.}$$

Ref: MK Bk. 5 Pg 233 – 235

WK. 5

PROFIT, LOSS, BUYING PRICE AND SELLING PRICE

Buying price (B.P) also termed as cost price (C.P)

LESSON 3

1. Finding the Cost price when profit and selling price are given.

Example

Nabisere sold a radio at shs. 50,000. She made a profit of sh. 10,000. What was her cost price?

$$C.P = S.P - \text{Profit}$$

$$C.P = 50,000 - 10,000$$

$$= \text{Shs. } 40,000$$

LESSON 4

2. Finding cost price when loss is given.

Oketch sold a goat at shs. 15,000. He made a loss of 3,000. How much did he buy it.

$$B.P = S.P + \text{Loss}$$

$$B.P = 15,000 + 30,000$$

$$B.P = 18,000$$

He bought the goat at shs. 18,000

LESSON 5

3. Finding selling price when profit and cost price are given.

A trader bought a shirt at sh. 7500, she sold it and made a profit of sh. 3500. What was the selling price?

$$S.P = B.P + \text{Profit}$$

$$S.P = \text{Sh. } 7,500 + \text{sh. } 3,500$$

$$S.P = \text{Shs. } 11,000$$

LESSON 6

4. Finding selling price when loss is given.

A pupil bought a ball at sh. 15,000 sold it and made a loss of sh. 3,000. What was the selling price of the ball.

$$S.P = B.P - \text{Loss}$$

$$S.P = \text{Sh. } 15,000 - 3000$$

$$S.P = \text{Shs. } 12,000$$

Reference : MK BK. 5 PG 245 – 249

WK. 6 LESSON 1

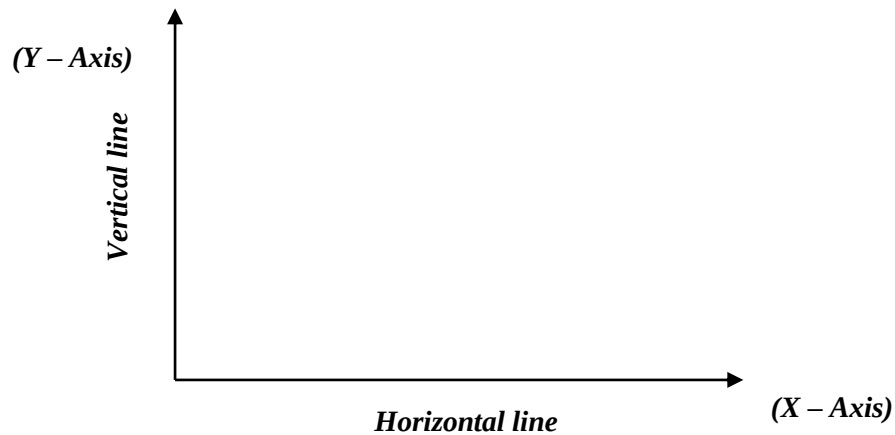
INTERPRETATION OF GRAPHS AND DATA.

Graphs (Features of graphs)

Graphs are used to represent collected information. We have different types of graphs. A good graph should have the following features.

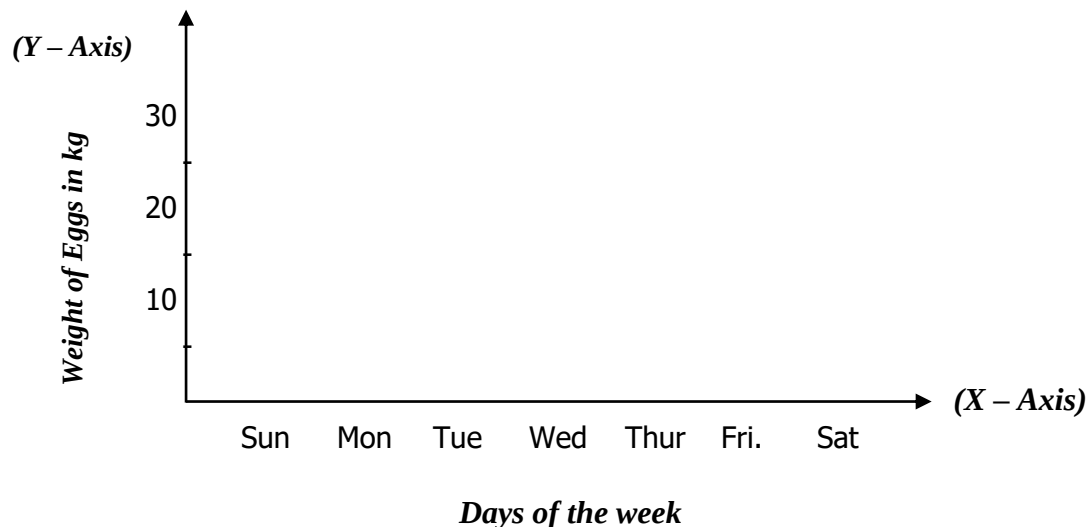
- (i) TITLE: which tells us what the graph is all about.
- (ii) THE VERTICAL AXIS : is a line that runs from top to bottom on the extreme left it is also called the Y – axis.
- (iii) The HORIZONTAL AXIS: is a line which runs from left to right on the extreme bottom. It is also called the X – axis
- (iv) THE SCALE: Both the Y – Axis and X – Axis of intervals suitable to scale.

Example I:



Each axis represents a different item being compared against the other. It may be **speed** against **time**, **days of the week** against **attendance**.

Example II



Each axis must be marked clearly showing items or variables. In the example above, the Y-axis represents the weight of eggs in kg and the scale of each interval used is 5kg. The x-axis represents days of the week where each interval stands for a day.

Ref: Fountain Maths Bk. 5 Pg. 230

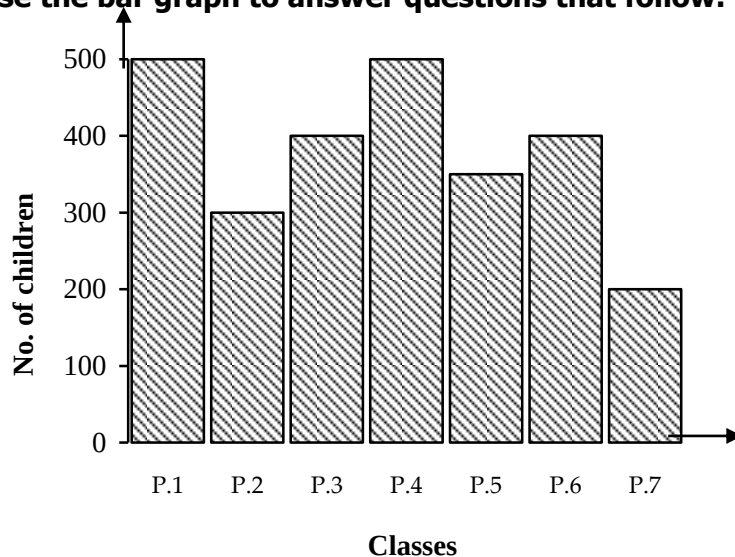
WK. 6 LESSON 2

INTERPRETATION OF BAR GRAPHS.

Collected information can be organized and represented on block or column structures forming a graph. This type of graph is called **a Bar graph**. The blocks or columns may lie horizontally forming a horizontal bar graph or upright forming upright bar graph.

Example

Use the bar graph to answer questions that follow.



Question 4

- Which class the least number of children?
- How many children are in P.6?
- How many more children are in P.4 than P.5
- Find the total number of children in the school.
- Find the range of pupils in the school.
- Find the average number of pupils per class.

Ref: Mk Bk. 5 (old edition) Pg. 259 - 262

WK. 6 LESSON 3

Drawing bar graphs

Example

The P.5 pupils counted vehicles which passed along the road near their school. The pupils recorded the results in a table as shown below. Use the information to construct a bar graph.

Type	Cars	Buses	Lorries	Motorcycle	Mini buses	
Total	12	8	6	10	16	

Suggest a suitable scale on the y – axis and along the x – axis. Decide which item will be on the y-axis and which one to be on the x – axis.

Scale

Let one small square represent 2 vehicles on the y – axis (vertical scale)

one small square represents one type of vehicle on the x – axis (Horizontal scale)

Scale reading on the Y – axis

Since one square represents 2 vehicles

12 can represent $\frac{12}{2} = 6$ squares.

16 minibuses = $\frac{16}{2} = 8$ squares.

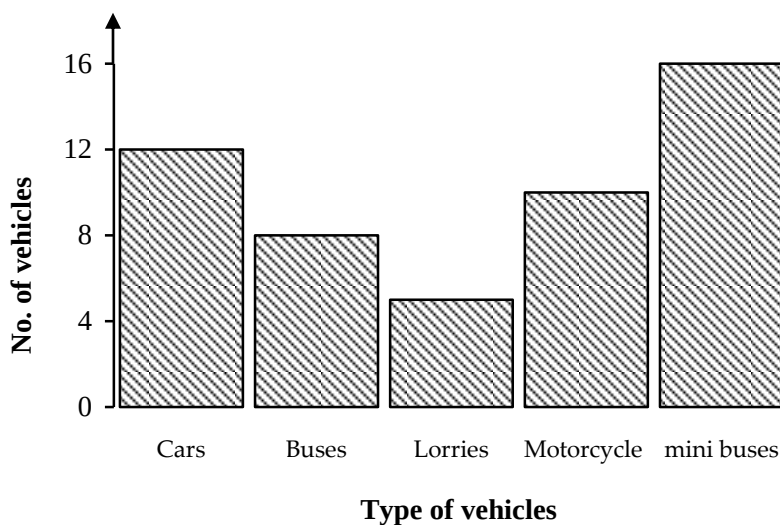
8 buses represent $\frac{8}{2} = 4$ squares.

10 motorcycles $\frac{10}{2} = 5$ squares.

6 lorries represent $\frac{6}{2} = 3$ squares.

Example

Use the bar graph to answer questions that follow.



WK.6 LESSON 4, 5 & 6

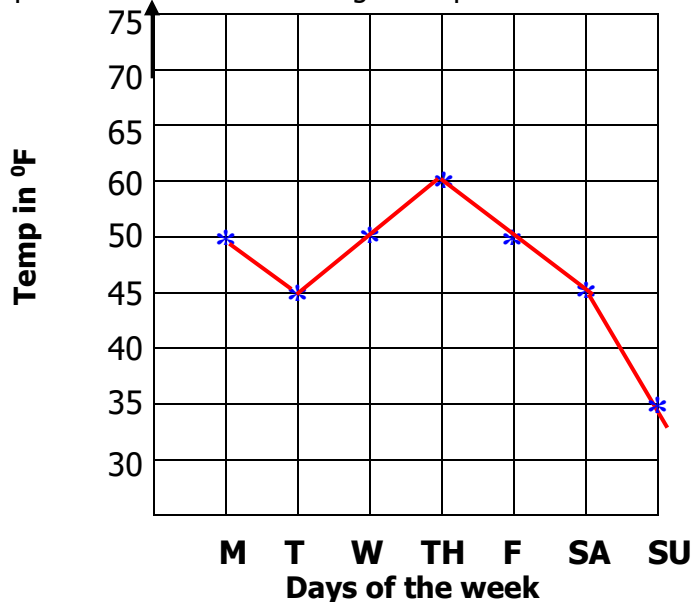
LINE GRAPHS

A line graph is where the information plotted on a graph is marked and the required point is indicated with a star, thereafter joined by a thick line.

Line graphs display information or data that change continuously over time.

Example

The graph below shows the midnight temperatures for the days of the last week.



Note:

- Which day had the lowest midnight temperature?
- Give three days which had the same midnight temperatures.
- What was the midnight temperature of Thursday?

Reference: MK BK. 5 Pg 267 – 268

Fountain & Math Bk. 5 pg. 240 - 242

WK. 7 LESSON 1 & 2

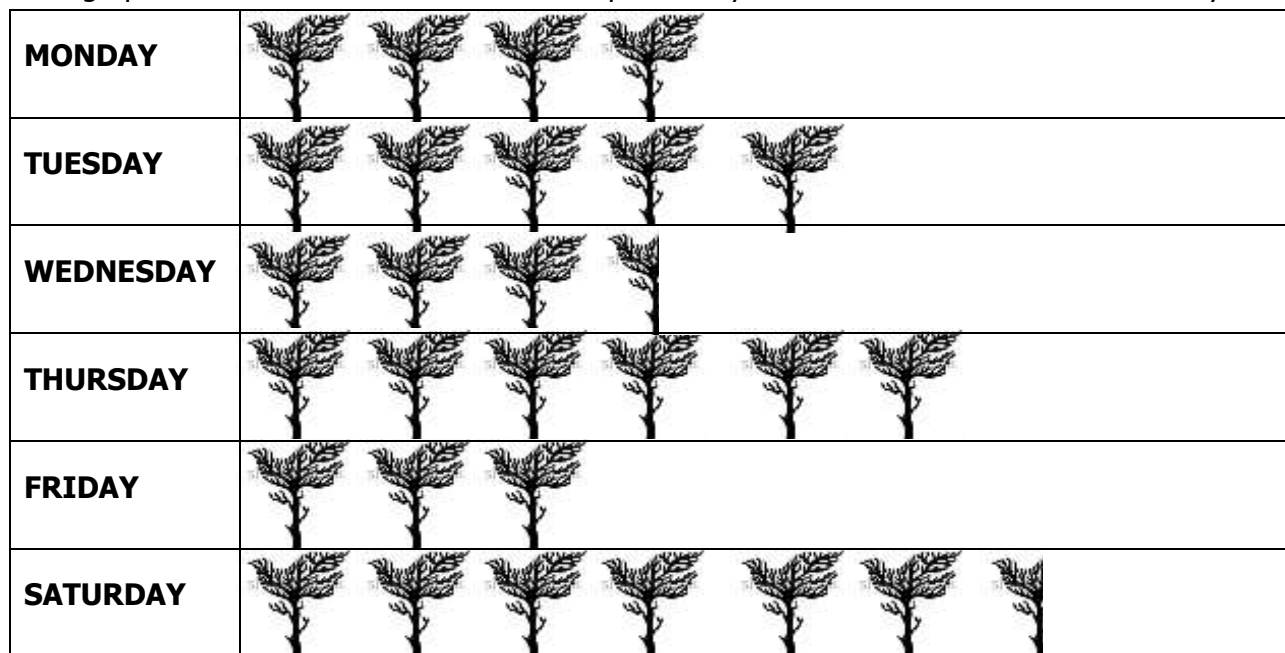
PICTOGRAPHS

These are graphs that use pictures or symbols to represent collected information.

The picture graph must have a scale in order to interpret the information accurately

Example

The graph below shows the number of trees planted by a farmer in Sironko district each day.



One  represents 50 trees

Question

- a) When were most trees planted?
- b) How many trees were planted on Tuesday?
- c) How many trees were planted on Saturday than Thursday?
- d) On which day did he plant the least number of trees?
- e) What was the range of trees planted in the whole week?
- f) Find the total number of trees planted in the whole week.

Ref: MK BK. 5 Pg 255 -257

Fountain Primary Math Bk. 5 pg. 243 – 245

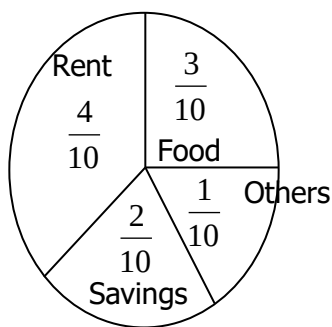
WK. 7 LESSON 3 & 4

PIE CHARTS

A pie chart or circle graph is a circular chart divided into sectors illustrating proportions. The sectors are divided by lines coming from the centre.

Example 1

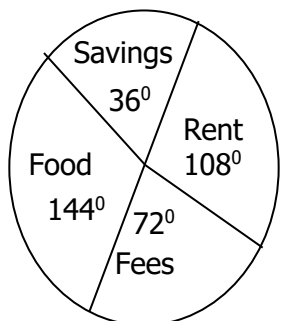
The pie chart below how a man spends shs. 30,000.



- i) How much does he spend on food?
- ii) How much does he spend on rent?
- iii) How much does he spend on other items?
- iv) Change the fraction for food to degrees.

Example 2

The pie chart below represents how a man spends sh. 12,000 in a month.



- i) How much does he spend on rent?
- ii) How much does he save?
- iii) What fraction represents fees?

Activity: BK. 6 MK. Pg. 180 ex. 8:12 No. 1, 3, 4 and 5